BMED 712 – Track A (Data-Driven Rehab ML): Five Projects

Dr. Mohamed Elgendi

Khalifa University, College of Medicine and Health Sciences

Spring 2026

Track A Project 1 – Robust Gait Phenotyping Across Pathologies (IMU)

- Clinically annotated multi-pathology gait dataset with **multi-IMU** recordings.
- Typical characteristics: multiple conditions, repeated trials, standardized walking protocols.
- **Dataset link:** Figshare – Multi-pathology IMU Gait Dataset

Core research question

- How robust are gait ML models across **pathologies**, **conditions**, and **sensor availability**?

Tasks (choose 2 required + 1 stretch)

- Required: pathology / impairment classification (multi-class).
- Required: **robustness benchmarking** (leave-one-group-out: subject / condition / pathology).
- Stretch: gait quality scoring or explainability (which sensors/axes matter most?).

Deliverables (paper-form report)

- Reproducible pipeline: preprocessing, splits, baselines, metrics, ablations.
- Ablations: **sensor ablation** (1 IMU vs 2 vs all), window size, normalization.
- "Error modes" section: where/why the model fails (clinically meaningful narrative).

Track A Project 2 – Exercise Quality & Technique Deviation Detection (IMU + MoCap)

- Rehab-focused dataset with **correct vs incorrect** movement variants.
- Modalities: IMU and/or MoCap (joint kinematics).
- **Dataset link:** GAITEX Dataset (IMU + MoCap)

Core research question

- Can ML detect **clinically meaningful deviations** (not just “which exercise”)?

Tasks

- Required: correct vs incorrect execution detection.
- Required: deviation type classification (or clustering + mapping if labels limited).
- Stretch: temporal localization (when the deviation occurs within the rep).

Deliverables (paper-form report)

- Baselines: classical features + shallow ML vs deep time-series model.
- Ablations: modality ablation (IMU only vs MoCap only vs fusion).
- “Clinical relevance mapping”: which deviations matter most and why.

Track A Project 3 – Speed-Invariant Rehabilitation Exercise Recognition (Multimodal)

- Rehab exercise dataset with **multiple speeds** and repeated reps.
- Modalities may include depth/vision, IMU, skeleton, or EMG (dataset-dependent).
- **Dataset link:** UL-RED – University of Liverpool Rehabilitation Exercise Dataset

Core research question

- How do models fail under **speed variability**, and how can we build **speed-invariant** predictors?

Tasks

- Required: exercise classification.
- Required: cross-speed generalization (train on normal speed; test on slow/fast).
- Stretch: rep counting / segmentation or self-supervised pretraining.

Deliverables (paper-form report)

- Benchmarks: within-speed vs cross-speed performance gap.
- Ablations: augmentation (time-warp, scaling) vs representation learning.
- Practical section: what robustness is required for home rehab deployment?

Track A Project 4 – Multimodal Gait Assessment with Vision + Inertial Data (Assistive Context)

- Dataset capturing posture/gait during assisted walking (e.g., smart walker context).
- Modalities: multi-camera / depth + inertial / MoCap (depending on dataset).
- **Dataset link:** PhysioNet – A multi-camera and multimodal dataset for posture and gait analysis

Core research question

- Does **multimodal sensing** meaningfully outperform single-modality models for rehab gait assessment?

Tasks

- Required: gait phase or event detection (heel-strike/toe-off) OR posture stability estimation.
- Required: modality benchmarking (vision-only vs inertial-only vs fusion).
- Stretch: missing-modality robustness (simulate camera dropout / IMU dropout).

Deliverables (paper-form report)

- Ablation suite: modalities, synchronization strategy, window length.
- Cost–benefit analysis: performance vs sensor complexity (deployment lens).
- "Failure taxonomy": occlusion, drift, latency, missing streams.

Track A Project 5 – Frailty-Aware Exercise & Physiological Response Modeling (ECG + IMU)

- Frailty or aging-related exercise tests with **wearable ECG** + **tri-axial acceleration**.
- Provides labels such as protocol phases, outcomes, or frailty groupings (dataset-dependent).
- **Dataset link:** PhysioNet – Wearable Exercise & Frailty Dataset

Core research question

- Can combining **movement + physiology** improve rehabilitation assessment and stratification?

Tasks

- Required: activity / phase classification (protocol segments).
- Required: physiological response profiling (HR/HRV features vs time + motion intensity).
- Stretch: group stratification (frailty-aware modeling) or fatigue detection proxy.

Deliverables (paper-form report)

- Baselines: motion-only vs ECG-only vs fusion.
- Ablations: signal quality control, feature sets, model family comparisons.
- Interpretation: link results to aging/frailty relevance + limitations section (ethics, bias).